

cluster_functional_types.R

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Cluster Taxa into Functional Types

Description

Assigns taxa into functional types (FTs) using a pre-computed Gower dissimilarity matrix, a pre-fitted hierarchical clustering object, and a predetermined number of groups ‘number_of_ft_groups’. The number of groups is expected to have been chosen upstream (e.g. via ‘select_ft_groups_by_silhouette()’).

Usage

```
cluster_functional_types(  
  data,  
  dist_mat,  
  hclust_obj,  
  number_of_ft_groups,  
  taxon_col = "taxon_name",  
  verbose = TRUE  
)
```

Arguments

data	A data frame with one row per taxon. Must contain a character column (identified by ‘taxon_col’) holding taxon names. Must have at least 4 rows and ‘nrow(data)’ must be strictly greater than ‘number_of_ft_groups’.
dist_mat	A ‘dist’ object produced by ‘compute_dissimilarity_matrix()’. Must inherit class “dist”. All distance values must be finite.
hclust_obj	An ‘hclust’ object produced by ‘fit_hclust()’. Must inherit class “hclust”.
number_of_ft_groups	A single positive integer giving the number of functional-type groups to cut the dendrogram at. Must be at least 2 and less than ‘nrow(data)’.

<code>taxon_col</code>	A single character string naming the column in 'data' that contains taxon names. Default: "taxon_name".
<code>verbose</code>	A single logical. If 'TRUE' (default), prints the chosen 'number_of_ft_groups' via 'cli::cli_inform()'.

Details

****Algorithm**:**

1. Cut the dendrogram at 'number_of_ft_groups' via 'stats::cutree()'.
2. Compute per-taxon silhouette widths via 'cluster::silhouette()'.
3. Return a tibble of taxon names, cluster assignments, and silhouette widths.

Distance computation (Inf -> NA, daisy, NaN -> 1.0) and hierarchical clustering are handled upstream by 'compute_dissimilarity_matrix()' and 'fit_hclust()' respectively. FT-groups selection is handled upstream by 'select_ft_groups_by_silhouette()'. If 'nrow(data) < 4' or 'number_of_ft_groups >= nrow(data)', the function aborts with an informative error.

Value

A tibble with columns:

taxon_name Taxon names (same values as 'data[[taxon_col]]'). Column name is always "taxon_name" regardless of 'taxon_col'.

functional_type Integer label (1..number_of_ft_groups) giving the functional-type cluster assignment for each taxon.

silhouette_width Per-taxon silhouette width for the chosen number_of_ft_groups solution.

See Also

[compute_dissimilarity_matrix()], [fit_hclust()], [select_ft_groups_by_silhouette()]

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